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**YISHUN JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION 2018**

BIOLOGY

9744/03

HIGHER 2

12 SEPTEMBER 2018

Paper 3 Long Structured and Free-response Questions

Wed 0800 - 1000

2 hours

Additional Materials: Writing papers

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READ THESE INSTRUCTIONS FIRST

Write your name and CTG in the spaces at the top of this page and on all separate answer paper used.

Write in dark blue or black pen only.

You may use a soft pencil for any diagrams or graphs.

Do not use paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **one** question on the separate answer papers provided.

The use of an approved scientific calculator is expected, where Appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE	
Section A	
Q1	/30
Q2	/10
Q3	/ 10
Section B	
Q4 / Q5	/25
Overall	/ 75

This document consists of **15** printed pages and **3** blank pages.

Section A

Answer **all** questions.

- 1 The Human Immunodeficiency Virus (HIV) infects helper T lymphocytes, including memory helper T lymphocytes, thereby weakening the immune system, causing the patient to succumb to secondary infections, leading to the development of acquired immunodeficiency diseases (AIDS).

Fig. 1.1 shows the binding of a HIV particle to a T lymphocyte. The viral receptor gp120 trimer binds to both CD4 and a co-receptor called CCR5 on the T lymphocyte surface.

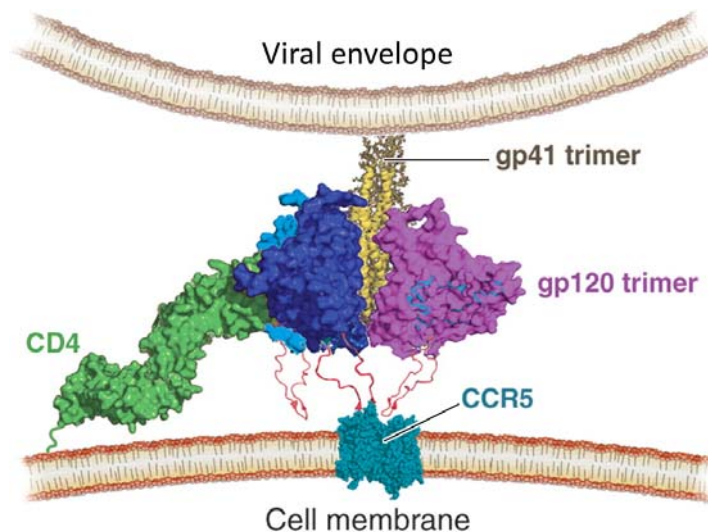


Fig. 1.1

- (a) Individuals who carry a mutation known as CCR5- Δ 32 in both copies of the CCR5 gene are resistant to certain strains of HIV. The CCR5- Δ 32 allele is a result of a deletion of a particular sequence of 32 base-pairs within the exon of CCR5 gene.

Explain how an individual who is homozygous for the CCR5- Δ 32 allele is resistant to HIV.

1. Loss-of-function mutation;
2. The deletion leads to a conformational change in the CCR5 co-receptor / gp120 binding domain of CCR5;
3. The shape of the altered CCR5 co-receptor no longer complementary to the shape of gp120 hence cannot bind;
4. Individual is homozygous – hence no normal CCR5 co-receptor is produced at all.
5. Entry of HIV is inhibited.

OR

6. Loss-of-function mutation;
7. The deletion results in the formation of a CCR5 polypeptide that will be mis-folded, leading to its degradation;
8. Absence of CCR5 co-receptor on the cell surface membrane;
9. Individual is homozygous – hence no normal CCR5 co-receptor is produced at all.
10. Entry of HIV is inhibited. [4]

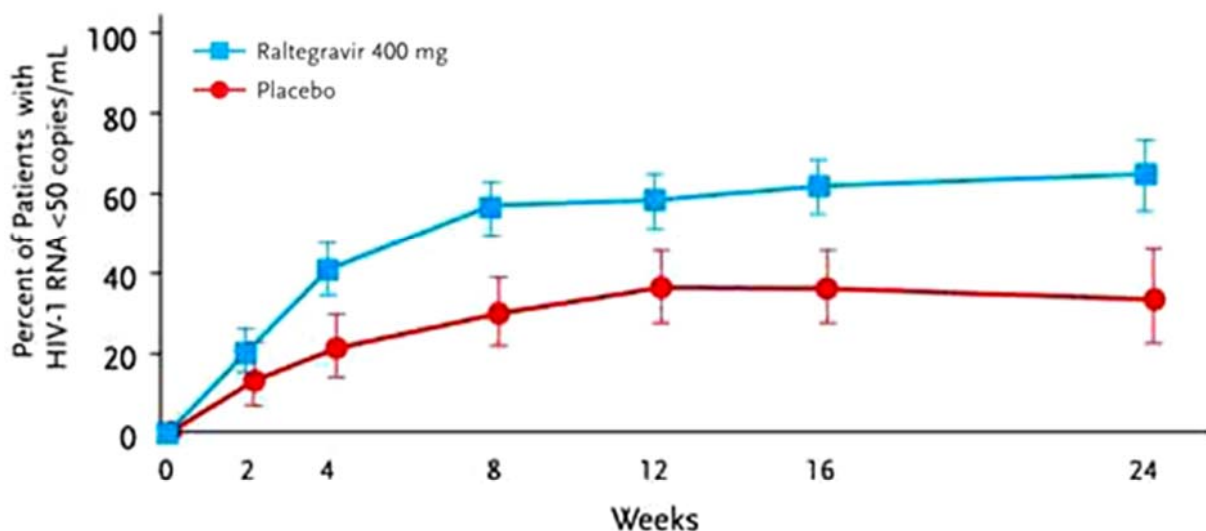
- (b) The allele frequency of CCR5- Δ 32 is unusually high in the western European countries, with up to 20% of ethnic western Europeans carrying this allele, which is rare or absent in all other ethnic groups throughout the world.

Suggest a reason for this phenomenon.

- Founders' effect – a small group of individuals from a larger population, some of whom carry the CCR5- Δ 32 allele, colonized/established a new population in the western European countries. [1]

There are two types of HIV: HIV-1 and HIV-2. HIV-1 integrase binds to specific sites at the ends of the viral DNA to facilitate its integration into the host genome. This allows the virus to persist in the latent stage for long periods of time.

Many integrase inhibitors have been discovered during the past 10 years, with some of them presently in clinical trials. An example of one such inhibitor is Raltegravir. In one such trial, one group of patients was given Raltegravir while the control group was given placebos (drug-free pills) over 24 weeks. The number of copies of HIV-1 RNA per ml of blood plasma was measured and the results are shown in Fig. 1.2.



Adapted from Merck & Co., Inc. Results of BENCHMRK-1 and-2, two phase-III studies evaluating the efficacy and safety of raltegravir, a novel HIV-1 integrase inhibitor, in patients with triple-class resistant virus. 14th Conference on Retroviruses and Opportunistic Infections; February 25–28, 2007; Los Angeles, CA.

Fig. 1.2

(c) With reference to Fig. 1.2, describe the effect of treating patients with 400mg of Raltegravir.

1. The groups of patients treated with Raltegravir had a higher percentage of patients showing HIV-1 RNA < 50 copies per ml as compared to the control group.
2. The percentage of patients with **HIV-1 RNA < 50 copies /ml increased from 20% in week 2 to 60% in week 24 in the group undergoing treatment whereas the percentage of patients with HIV-1 RNA < 50 copies /ml in the control group increased from 10% in week 2 to 30% in week 24.** [2]

The onset of AIDS, which can occur many years later, coincides with a severely lowered primary and secondary immune response, owing to greatly reduced numbers of helper T lymphocytes in the body.

(d) Explain how the immune system defends against HIV infection.

1. Helper T cells that are infected by HIV are recognised by natural killer cells which induce the infected cells to undergo apoptosis;
2. Antigen-presenting cells (APCs) (macrophages/dendritic cells) take up /engulf the virus by phagocytosis;
3. the viral antigens are processed into short peptides / epitopes and presented on (MHC complexes) on their cell surfaces;
4. APCs secrete cytokines which activate naïve T lymphocytes/cells which undergo clonal expansion and differentiation into effector T cells (such as helper T cells and cytotoxic T cells) and memory T cells;
5. Cytotoxic T cells recognise antigens on the surface of infected helper T cells via the T cell receptor and induce them to undergo apoptosis;
6. As most helper T cells are infected by HIV, they cannot activate naïve B lymphocytes/cells to undergo clonal expansion and differentiation to form plasma cells;
7. hence little/no antibodies are produced against the virus;

OR

Some helper T cells that are not infected by HIV can activate naïve B lymphocytes, which undergo clonal expansion and differentiation into plasma cells and memory B cells. Plasma cells can secrete antibodies against HIV. [4]

- (e) Most T lymphocyte development occurs in an organ called the thymus. As part of this development, DNA sequences are cut out from the T lymphocyte genes, forming T lymphocyte receptor excision circles (TRECs).

The number of TRECs is thought to correlate with thymus function. CD4⁺ progenitor cells from individuals progressing to AIDS have a **marked loss in T lymphocyte development capacity**. The presence of TRECs in T lymphocytes can be detected in blood samples of HIV-infected patients using PCR.

- (i) Suggest why PCR is required in order to identify the presence of TRECs in the blood samples.

1. **Marked loss in T lymphocyte development capacity will lead to a significant reduction in the number of TRECs in HIV-infected individuals;**
2. **PCR is required to amplify the amount of TRECs extracted from blood samples of HIV-infected patients.** [2]

- (ii) Predict the amount of TRECs present in patients undergoing highly active anti-retroviral therapy (HAART) and give a reason for your answer.

1. **Administration of HAART to HIV-infected patients can prevent HIV from replicating and reduce the viral load;**
2. **resulting in the restoration / enhancement of CD4⁺ cell count, ∴ increasing the TREC content / amount.** [2]

- (f) A person who is confirmed as HIV-positive was tested positive for the presence of antibodies to HIV. A human can make more than 10^{12} different antibody molecules as a means to defend against potential pathogens such as the HIV.

Explain how a large diversity of antibodies with different specificity is produced.

1. **The specificity of antibodies depends on the variable regions which are encoded by the variable, joining, and diversity gene segments, each of which are present in multiple copies in the genome;**
2. **Somatic recombination occurs during **B cell maturation** in bone marrow where there is DNA rearrangement;**
3. **at Ig heavy chain gene locus where one V segment, one D segment and one J segment are selected to form a single VDJ exon;**
4. **at Ig light chain gene locus where one V segment and one J segment are selected to form a single VJ exon;**
5. **leading to **combinatorial diversity**;**
6. ****Combinatorial diversity** is also created from the association of different light and heavy chains to form an antibody;**
7. **Somatic hyper-mutation which is a random point mutation in the rearranged VDJ / VJ regions during **activation** of naïve B cells.** [4]

In July 1990, a young woman in Florida, who had no known risk factors for HIV infection and no known contact with other HIV-positive persons, was tested HIV positive after undergoing an invasive dental procedure performed by a dentist who had AIDS.

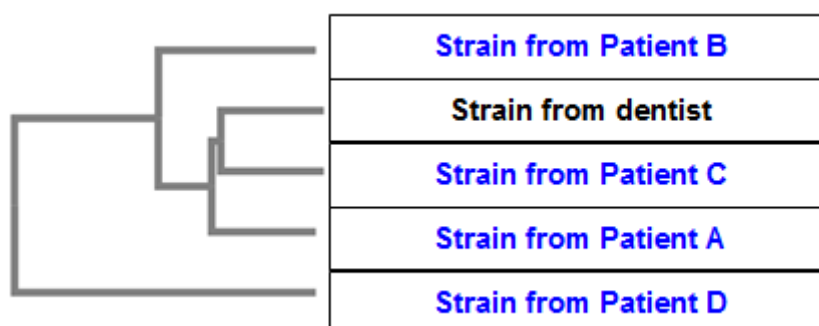
The U.S. Centre for Disease Control and Prevention (CDC) carried out an epidemiological investigation using DNA isolated from white blood cells of the dentist and patients A to D. The DNA was sequenced and compared with the strain isolated from the dentist.

Fig. 1.3 shows the multiple sequence alignment before a phylogenetic analysis could be carried out.

Strain from	Sequence																			
Dentist	-	-	-	-	C	-	T	A	-	T	T	G	-	C	T	G	G	C	G	C
Patient A	-	-	G	-	C	-	C	A	-	T	A	G	-	C	T	A	G	C	G	C
Patient B	-	-	G	-	C	A	C	C	-	T	-	G	-	C	T	A	G	C	G	C
Patient C	-	-	G	-	C	-	T	-	-	T	G	G	G	C	T	G	G	C	G	C
Patient D	C	A	G	A	C	-	T	A	C	T	-	G	-	C	T	A	G	-	G	-

Fig. 1.3

- (h) (i) Complete the figure below to show how closely related the different strains of HIV are.



[2]

- (ii) Explain how you derive your answer in (h) (i).

- HIV strain from patient C has the least number of nucleotide differences / has 3 nucleotide differences from the proviral DNA extracted compared to the strain from the dentist;
- fewer nucleotide differences indicate that the 2 viral strains share a more recent common ancestor and have diverged more recently from their common ancestor;

[2]

Viruses are susceptible to the forces of natural selection. Evolutionary changes in the pathogenesis of the HIV have been instrumental in the transmissibility of the virus among people.

Fig. 1.4a and Fig. 1.4b below show the general trend of **HIV pathogenesis** against the **transmissibility of HIV**, and the **number of infected individuals over time**, respectively.

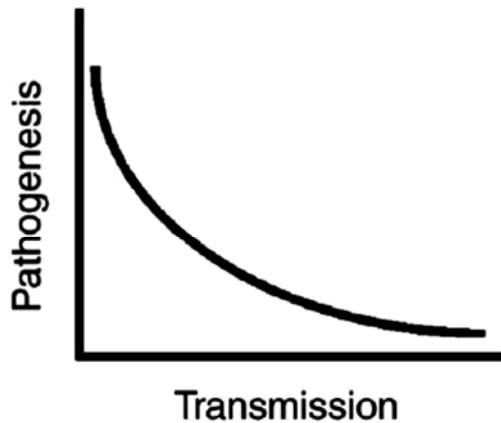


Fig. 1.4a

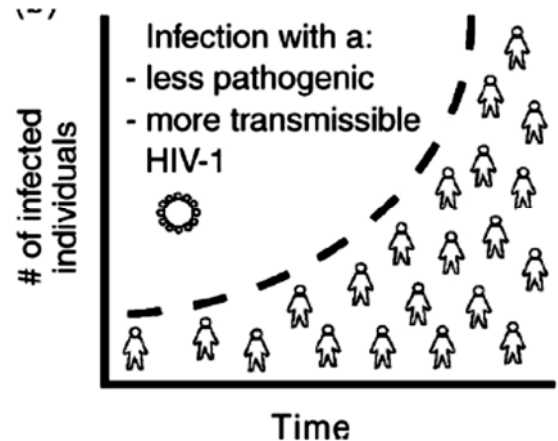


Fig. 1.4b

(i) Explain how the observed trends in Fig. 1.4(a) and (b) support Darwin's theory of natural selection.

1. There is variation in the pathogenesis of HIV strains;
2. HIV strains that are less pathogenic will result in a greater number of transmissibility among individuals, indicated by the greater number of infected individuals over time;
3. These HIV strains are at a selective advantage / are selected for, and are thus more able to replicate and pass down their alleles;
4. Over many generations, HIV strains that are less pathogenic will dominate the population. [4]

Bacteria has a natural defence system against viral infections involving RNA sequences known as clustered regularly interspaced short palindromic repeats (CRISPR).

When the bacteriophage infects a bacterial cell, the viral genome released into the bacterial cell is cleaved. Subsequently, a cleaved portion of the viral genome is integrated into the bacterial genome. The bacterial cell detects the phage DNA integrated within its genome and produces a type of RNA known as the CRISPR RNA. The CRISPR RNA contains a sequence that is complementary to that of the integrated phage DNA. When the next phage infects the same bacterial cell, the CRISPR RNA will bind to its target sequence in the viral genome and a nuclease is recruited to cut the phage DNA, disabling the invading phage.

The sequence of the CRISPR RNA can be edited and subsequently used to cut any DNA sequence at a precisely chosen site in eukaryotic cells. This gives rise to the possibility of correcting mutations associated with genetic disorders. Fig. 1.5 shows how the sequence of a defective allele can be replaced with the sequence of a normal allele from a donor DNA via homologous recombination using the CRISPR technology.

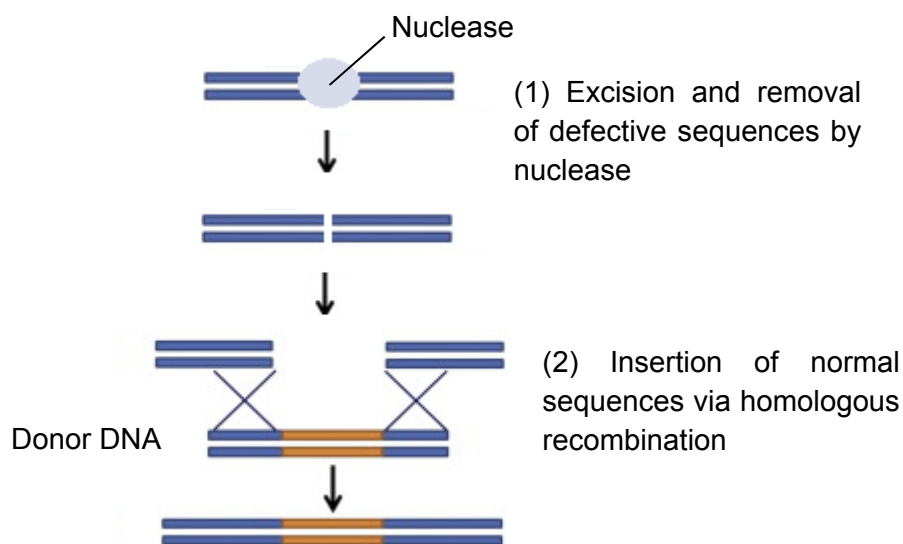


Fig. 1.5

- (j) (i) Another technology that can be used to treat genetic disorders is gene therapy. Similar to the CRISPR technology, gene therapy involves the introduction of normal alleles into patients. However, it does not involve the excision and removal of defective alleles. Hence, it can only be used to treat recessive genetic disorders.

With reference to Fig. 1.5, explain the advantages of using CRISPR technology to treat genetic disorders over gene therapy.

1. CRISPR technology can be used to treat both dominant and recessive genetic disorders;
2. as nuclease creates a double stranded break in the DNA to remove the dominant alleles (and gene function is restored when normal alleles are inserted via homologous recombination);
3. However, insertion of normal alleles in gene therapy is unable to treat dominant genetic disorders as only a copy of dominant allele is required to show its effect in patients;

4. Normal alleles introduced during gene therapy can be degraded overtime/ temporary/ may not be integrated in the genome vs. long lasting treatment/ one treatment required for CRISPR;
5. CRISPR successfully correct the genetic disorder such that subsequent generations of cells are normal while gene therapy can only affect one generation of cells;
6. Gene therapy has a risk of insertional mutagenesis (if retroviruses are used as a vector) while CRISPR does not have this risk;
7. as in gene therapy, insertion is not specific while in CRISPR, insertion is specific/ precise; [2]

- (ii) The successful late transcription of HIV provirus following viral activation is highly dependent on the early expression of the viral regulatory proteins Tat and Rev. The **elongation of viral mRNA** from the integrated provirus is **initiated by Tat**, while the **nuclear export of viral mRNA transcripts** is **regulated by Rev**. In **HIV-infected activated T lymphocytes**, the combination of Tat and Rev provide a **very high level of viral gene expression**, while the same proteins in **resting T lymphocytes** are important for **maintaining the provirus in a latent state**.

The CRISPR system has been shown in research to **abolish Tat and Rev protein expression**.

Suggest how the CRISPR system can potentially be used to suppress viral replication.

1. With the abolishment of Tat and Rev protein expression, viral gene expression may be successfully diminished in HIV-infected activated T lymphocytes, preventing successful viral replication / suppressing viral replication; [1]

[Total: 30]

- 2 An investigation was carried out into the response of sorghum being kept at a low temperature for a short period of time. Soybean plants, which are better adapted than sorghum for growth in subtropical and temperate climates, were used for comparison.

Plants of sorghum and soybean were kept at 25°C for several weeks and then at 10°C for three days. The temperature was then increased to 25°C again for seven days. Day length, light intensity and carbon dioxide concentration were kept constant throughout.

The uptake of carbon dioxide, as mg CO₂ absorbed per gram of leaf dry mass, was measured

- at 25 °C before cooling;
- on each of the three days at 10°C;
- for seven days at 25°C. The results are shown in Table 2.1.

Table 2.1

plant	carbon dioxide uptake / mg CO ₂ g ⁻¹				
	at 25 °C, before cooling	at 10 °C			at 25 °C (mean over days 4 to 10)
		day 1	day 2	day 3	
sorghum	48.2	5.5	2.9	1.2	1.5
soybean	23.2	5.2	3.1	1.6	6.4

- (a) Compare the changes in carbon dioxide uptake in sorghum and soybean during the three days at 10°C. [2]

1. greater reduction in carbon dioxide uptake sorghum than in soybean;
2. use of comparative figures:
E.g. sorghum: reduction in carbon dioxide uptake from 5.5 to 1.2 mg CO₂ g⁻¹ or by 4.3 mg CO₂ g⁻¹ while soybean: reduction in carbon dioxide uptake from 5.2 to 1.6 mg CO₂ g⁻¹ or by 3.6 mg CO₂ g⁻¹

During the cooling period, the ultrastructure of the sorghum chloroplasts changed. The membranes of the thylakoids moved closer together, eliminating the spaces between them. The size and number of grana became reduced.

(b) Explain how these changes could be responsible for the low rate of carbon dioxide uptake by sorghum plants even when they are returned to a temperature of 25°C. [4]

1. less surface area ;
2. less absorption of light ;
3. less, photophosphorylation / light dependent reaction ;
4. less chemiosmosis ;
5. (due to) smaller thylakoid space or reduced proton gradient ;
6. less ATP (produced) ;
7. less reduced NADP (produced) ;
8. light-independent reaction / Calvin cycle, slows down ;
9. less carbon dioxide, fixed / combined with PEP ; R uptake

Fig. 2.1 shows the changes in concentration of a 3C compound, glycerate phosphate, GP, and a 5C compound, ribulose biphosphate, RuBP, extracted from samples taken from actively photosynthesising green algae in an experimental chamber when the light source was turned off.

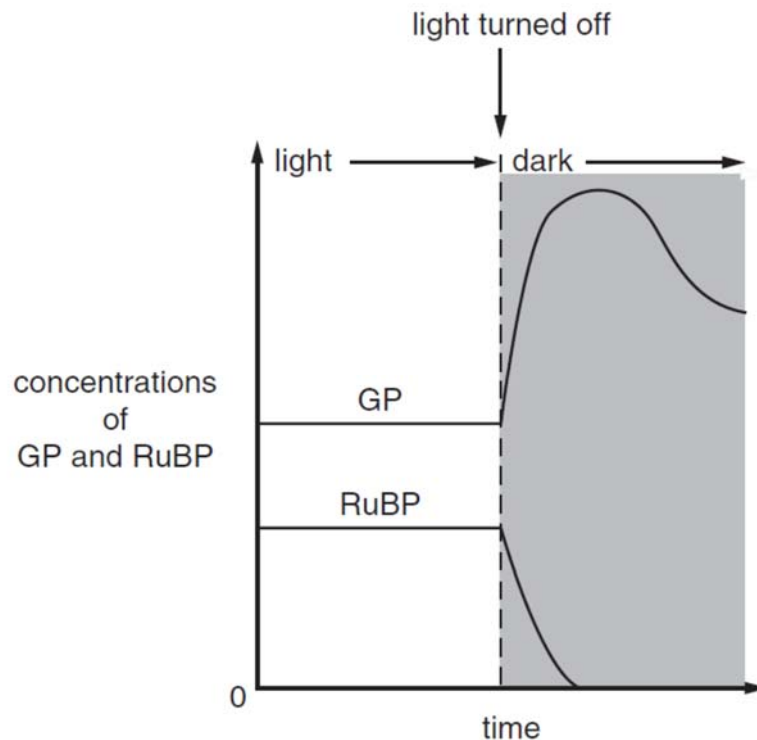


Fig. 2.1

- (c) With reference to Fig. 2.1, explain what happens to the concentration of GP and RuBP after the light source was turned off. [4]

GP [2]

- GP increase as continues to form RuBP;
- until all the RuBP is used up;
- GP falls as it is converted to hexose/ glucose

RuBP [2]

- In the dark, RuBP not regenerated;
- ATP and reduced NADP are required from light-dependent reactions/photophosphorylation

[Total: 10]

3 Dengue fever is an infectious disease transmitted by a vector.

(a) State the full name of the vector that transmits the pathogen.

Aedes aegypti ;; to be written as Aedes aegypti

(b) Global warming could lead to transmission of dengue fever beyond the tropics.

Outline how global warming could change the area of distribution of dengue fever.

1. Global warming results in increase in temperature;
2. The temperate regions become warmer, Mosquitoes move to higher latitude so they spread to new areas expanding their distribution;
3. Global warming also results in higher altitudes becoming warmer;
4. thus conditions become more suited for the survival of Aedes mosquitoes; /thus mosquitoes will be able to colonise altitudes higher up the mountain;
5. Global warming could result in more rainfall in certain areas;
6. creating more pools of stagnant water for mosquito breeding (ORA less rainfall in some areas so less standing water for mosquitoes to breed);

In 2018, the World Health Organization (WHO) estimated that 3.9 billion people were at risk of dengue fever.

Table 3.1 shows the number of cases of dengue fever and the number of deaths from dengue fever between 2000 to 2012, for all the countries of Asia and for Singapore.

The table also shows the numbers in Singapore as percentages of the numbers for all countries in Asia.

Table 3.1

Year	Number of cases of dengue fever ¹		Cases in Singapore as a percentage of Asia	Number of deaths from dengue fever ¹		Deaths in Singapore as a percentage of Asia
	Asia	Singapore		Asia	Singapore	
2012	353 210	4 632	1.31	1 248	2	0.16
2010	352 107	5 364	1.52	1 074	6	0.56
2008	206 493	7 032	3.41	668	0	0.00
2006	171 579	3 127	1.82	694	10	1.44
2004	158 848	9 459	5.95	568	8	1.41
2002	108 482	3 945	3.64	514	6	1.17
2000	46 562	673	1.45	247	2	0.81

¹ World Health Organisation (WHO) Western Pacific Region

(c) Describe the trends shown in Table 3.1.

1. numbers of cases have increased in Asia from 46 562 in 2000 to 353 210 in 2012;
2. numbers of cases in Singapore shows fluctuations with high incidences of 9459 in 2004 and 7032 in 2008 ;
3. number of deaths have increased in Asia from 247 in 2000 to 1248 in 2012 ;
4. number of deaths in Singapore remains low with high incidences of 8 and 10 in 2004 and 2006 respectively ;

- (c) Suggest reasons why the number of cases of dengue fever and the number of deaths from dengue fever changed between 2008 and 2012 in Singapore.

three from

1. example of control of breeding of, vector / mosquitoes;
e.g. drainage of stagnant water / sterile males using *Wolbachia* technology / aerial spraying of insecticide / oil on water / fish in water
2. example of reduction of contact between vector and humans ;
e.g. bed nets (impregnated with insecticide) / insect repellents
3. earlier, identification of cases / treatment of dengue fever ;
4. better, awareness of / education about, transmission / control methods ;
5. AVP ; e.g. targeting people at risk (e.g. pregnant women / better screening of blood for transfusion)

R *better access to, healthcare / AW, without further qualification*

[Total: 10]

Section B

Answer **one** question in this section.

Write your answers on separate writing papers provided.

Your answers should be illustrated in continuous prose, where appropriate.

Your answers must be set out in parts **(a)** and **(b)**, as indicated in the question.

Mark scheme abbreviations

; separates marking points

/ alternative answers for the same point

R reject

A accept (for answers correctly cued by the question, or by extra guidance)

AW alternative wording (where responses vary more than usual)

underline actual word given must be used by candidate (grammatical variants accepted)

max indicates the maximum number of marks that can be given

ora or reverse argument

mp marking point (with relevant number)

ecf error carried forward

I ignore

AVP alternative valid point

4 (a) Describe how the diversity of bacteria and virus is achieved.

[15]

Any 3 from:

1. Transformation, uptake of DNA (from the environment) **AW**;
2. By competent bacteria;
3. *ref to* entry of one strand **AW**
4. *ref to* homologous recombination / double cross-over **R** cross-over

Any 3 from:

5. Transduction, (bacteriophage) attaches to bacterial cell wall (receptor) and injects viral DNA into bacteria;
6. *ref to* generalised transduction; i.e. parts of degraded bacterial DNA repackaged into new transducing phages;
7. *ref to* specialised transduction; i.e. excise of prophage and part of bacteria DNA (during spontaneous induction), repackaging into new transducing phages;
8. (Resultant) transducing phages lyse the bacterial cell to infect other recipient bacteria;
9. *ref to* homologous recombination / double cross-over **R** cross-over

Any 3 from:

10. Conjugation between F⁺ cell and recipient F⁻ cell;
11. formation of sex pilus by F⁺ cells;
12. attachment of sex pilus to F⁻ cells;
13. formation of cytoplasmic mating bridge;
14. *ref to* transfer of F plasmid from the F⁺ cell to the F⁻ cell

Any 3 from: Antigenic shift

- 15. two or more strains of viruses infected the same host cells;
- 16. random reassortment of the virus genome;
- 17. new combination of virus genome;
- 18. encodes for different surface glycoproteins;

Any 3 from: Antigenic drift

- 19. lack of proof-reading ability of RNA-dependent RNA polymerase / reverse transcriptase
- 20. mutations of the virus genome;
- 21. mutations accumulate over time;
- 22. gradual shift in the coding of the surface glycoproteins;
- 23. AVP; e.g. (random) mutation, generates new alleles;

- (b) The structural genes in the bacterial operon can be regulated by either the inducible or repressible system.

Discuss the extent, to which the two systems of regulation can be similar and yet different. [10]

QWC: At least 1 similarity and 1 difference highlighted for balance view

Similarities

- 1. Regulate gene expression for proteins in the same metabolic pathway;
- 2. *Ref to regulation as a whole unit*;
- 3. *Ref to rapid response*;
- 4. *Ref to no wastage of energy / resources / ATP*;
AW e.g. gene products produced when needed;
- 5. Structural genes under control of a single promoter and operator;
- 6. *Ref to negative control by repressor*;
AW repressor bound to operator to block transcription;

Differences

- 7. Inducible, functions in catabolic pathway;
- 8. Repressible, functions in anabolic pathway;
- 9. Inducible, active repressor **A** repressor bound to operator;
- 10. Repressible, inactive repressor **A** repressor does not bind to operator;
- 11. Inducible, gene expression active in the presence of inducer;
- 12. Repressible, gene expression inactive in the presence of co-repressor **ORA**;
- 13. Inducible, inactivation of repressor by inducer;
- 14. Repressor, activation of repressor by co-repressor;
- 15. Inducible, operon active when substrates are present;
- 16. Repressible, operon inactive when products are excess;
- 17. Inducible, operon switched off by default;
- 18. Repressible, operon switched on by default;

I positive control / CAP / CAP binding site / cAMP

[Total: 25]

5 (a) Haematopoietic stem cells are found in the bone marrow.

Discuss how haematopoietic stem cells can give rise to naïve B-cells which upon activation, will form plasma cells that regulate the production of immunoglobulins to overcome pathogen infection. [15]

(1) haematopoietic stem cells can give rise to naïve B-cells **MAX 4**

1. *ref to* haematopoietic stem cells being multipotent;
2. differentiate to form different type of blood cells;
3. give rise to cells from both myeloid and lymphoid lineage;
4. lymphoid progenitor cell further differentiation to naïve B-cell;
5. *ref to* somatic recombination
6. *ref to* VDJ recombination for heavy chain
7. *ref to* VJ recombination for light chain
8. formation of specific B-cell receptors
9. clonal expansion and selection **AW** test for self-reactivity

(2) activation of naïve B-cell to form plasma cell **MAX 4**

1. recognition and binding of antigen by specific naïve B-cell receptors;
2. antigen complementary variable region of B-cell receptor;
3. *ref to* antigen presentation by B-cells on MHC II receptors;
4. *ref to* recognition of antigen by specific T_H-cells receptors;
5. activation of B-cells, differentiation of naïve B-cells to both plasma cells and memory cells **A** clonal expansion and differentiation of B-cells;

(3) regulation of production of immunoglobulins **MAX 4**

1. heavy chain gene and light chain gene in nucleus;
2. *ref to* transcription heavy chain gene and light chain gene;
3. *ref to* post-transcription modification **A** RNA splicing, additional of 7-methylguanosine at 5'-end, 3'-polyadenylation
4. *ref to* translation of mRNA by bound 80S ribosomes to ER membrane;
5. *ref to* assembly of heavy and light chains to form functional immunoglobulins;
6. *ref to* antibody having quaternary structure;
7. *ref to* antibody having 2 heavy and 2 light chains;
8. *ref to* antibody having variable and constant regions;
9. *ref to* class switching;
10. *ref to* somatic hypermutation;
11. *ref to* activator binding to enhancer to upregulate production of immunoglobulins **ORA**

(4) overcome pathogen infection **MAX 4**

1. phagocytosis by macrophages;
2. *ref to* antigen presentation by macrophages on MHC II receptors;
3. *ref to* activation of naïve T-cells;
4. *ref to* formation of T_H cells;
5. *ref to* roles of antibodies
 opsonisation;
 antibody-dependent cell toxicity;
 agglutination;
 activation of complement system;

QWC (1m) points communicated clearly without ambiguity and with relevant connections as to how antibody (protein) is expressed as HSC differentiate to form B-cell which is subsequently activated by exposure to antigen

(b) Describe the various bonds and their importance in carbohydrates.

[10]

Any nine from:

Formation of glycosidic bond

1. condensation reaction;
2. elimination of one water molecule;

$\alpha(1\rightarrow4)$ glycosidic bond

3. between hydroxyl group of carbon 1 and hydroxyl group of carbon 4;
4. found in starch / glycogen along the helical coil;
5. resulting in a more compact shape for storage;

$\alpha(1\rightarrow6)$ glycosidic bond

6. between hydroxyl group of carbon 1 and hydroxyl group of carbon 6;
7. found in amylopectin at branch point;
8. (5) resulting in a more compact shape for storage;
9. many branch ends allow a number of amylase to act on at any one time;
10. for easily broken down / hydrolysis of polymer;

hydrogen bond

11. intra-chain H-bonding between hydroxyl groups helps stabilise helical structure

$\beta(1\rightarrow4)$ glycosidic bond

12. β glucose, 180° rotation of alternating glucose residues;
13. forms linear structure of cellulose chain

hydrogen bond

14. Hydroxyl groups project outwards,
formation of hydrogen bonds between adjacent chains,
rigid cross-linking between the chains;
15. Microfibrils arranged in larger bundles to form macrofibrils.
great tensile strength

16. AVP;

QWC (1m): at least 1 points from each of the different bonds for full credit

[Total: 25]